

Contact Information	5500 Wabash Avenue Terre Haute, IN 47803	luddenig[AT] rose-hulman[DOT]edu
Education	University of Illinois Urbana-Champaign (Illinois) , Urbana, IL – Ph.D., Computer Science Advisor: Sheldon H. Jacobson Thesis: “Graph Partitioning: Redistricting Games & the Spherical Zoning Problem”	Aug 2023
	Rose-Hulman Institute of Technology (RHIT) , Terre Haute, IN – B.S., Computer Engineering & Mathematics	Nov 2016
Experience	Rose-Hulman Institute of Technology (RHIT) , Terre Haute, IN – Assistant Professor of Computer Science and Software Engineering	Aug 2023 – Present
Honors	NSF Graduate Research Fellow Outstanding Teaching Assistant–Lifetime , Illinois CS Finalist – <i>Research Live!</i> , Illinois Graduate College Graduate Teacher Certificate , Illinois CITL Mavis Future Faculty Fellow , Grainger College of Engineering Outstanding Teaching Assistant , Illinois CS Saburo Muroga Endowed Fellowship , Illinois CS	2019–2023 Spring 2022 Spring 2022 Spring 2021 2020–2021 Fall 2019 2017–2018
Teaching	Rose-Hulman Institute of Technology – CSSE/MA 474: Theory of Computation – CSSE/MA 490 (Special Topics): Algorithmic Game Theory – CSSE 230: Data Structures and Algorithm Analysis – CSSE 220: Object-Oriented Software Development – CSSE/MA 490 (Special Topics): Algorithmic Game Theory – CSSE/MA 473: Design and Analysis of Algorithms – CSSE/MA 473: Design and Analysis of Algorithms – CSSE 220: Object-Oriented Software Development – CSSE/MA 474: Theory of Computation – CSSE/MA 474: Theory of Computation – CSSE 220: Object-Oriented Software Development Instructor of Record, Illinois CS – CS 173: Discrete Structures (Section AL1, asynchronous online) Teaching Assistant, Illinois CS – CS 482/IE 413: Simulation – CS 482/IE 413: Simulation * CS 374: Algorithms & Models of Computation – CS 482/IE 413: Simulation – CS 481/IE 410: Stochastic Processes * CS 173: Discrete Structures * CS 173: Discrete Structures Course Aide, Grainger College of Engineering – ENG 598 TL: Teaching and Leadership * — Recognized in the CITL List of Teachers Ranked as Excellent	Spring 2026 Winter 2025–26 Winter 2025–26 Fall 2025 Spring 2025 Spring 2025 Winter 2024–25 Fall 2024 Spring 2024 Winter 2023–24 Fall 2023 Summer 2020 Spring 2021 Spring 2020 Fall 2019 Spring 2019 Fall 2018 Spring 2018 Fall 2017 Fall 2019 – Spring 2023

Service	Student Affairs Committee, RHIT	Sep 2024 – Present
	– Co-chair since Aug 2025	
	– Review applications for new student organizations and provide feedback	
	Integrity Committee, RHIT CSSE	Sep 2024 – Present
	– Review academic misconduct allegations and provide recommendations	
	– Develop departmental policies for defining and investigating misconduct	
	Freshman Mentoring Committee, RHIT CSSE	Aug 2024 – Present
	– Chair since Aug 2025	
	– Facilitate peer mentoring through group events and one-on-one mentoring	
	Process Committee, RHIT CSSE	Aug 2024 – Aug 2025
	– Refine continuous improvement process for ABET accreditation	
	– Review assessment rubrics for clarity and consistency	
	Social Media + Newsletter Committee, RHIT CSSE	Sep 2023 – Present
	– Collaborate with other committee members to write, edit, and distribute monthly Student Spotlight newsletter	
	Session Facilitator for RHIT August Teaching Workshop	Aug 2024
	– Plan & facilitate “Classroom Assessment” session for new/returning faculty	
	Application Reviewer for Noblitt Scholars Program	Jan 2024, 2025, 2026
	– Review applicant videos and materials as part of the selection process	
	Grad Fellow, CRA-E Committee	June 2020 – June 2022
	– Manage, write, and edit Undergraduate Research Highlights for CRA-E website	
	– Plan and deliver webinar for undergraduates considering a PhD in CS	
	– Provide graduate student perspective on CRA-E activities during annual meeting	
	Grad Academy for College Teaching Volunteer, Illinois CITL	Fall 2018 – Spring 2023
	– Facilitate pre-semester small-group session for new CS teaching assistants	
	Journal reviewing	2019 – Present
	– <i>The American Statistician</i>	– <i>Journal of Air Transport Management</i>
	– <i>Computational Opt. and Applications</i>	– <i>Journal of Computational Social Science</i>
	– <i>Computers and Operations Research</i>	– <i>Journal of Quantitative Analysis in Sports</i>
	– <i>Discrete Optimization</i>	– <i>Operations Research</i>
	– <i>Geographical Analysis</i>	– <i>Optimization Letters</i>
	– <i>INFORMS Journal on Computing</i>	– <i>Networks</i>

Peer-reviewed
Journal Papers

1. Deshpande, S., **I.G. Ludden**, and S.H. Jacobson (2025). "Bias in the ballot: how votemandering exploits gerrymandering and campaign strategies." *Annals of Operations Research*, 352:121–168. DOI: [10.1007/s10479-025-06748-9](#).
2. Dobbs, K.W., D.M. King, **I.G. Ludden**, and S.H. Jacobson (2024). "Facilitating Compromise in Redistricting with Transfer Distance Midpoints." *INFORMS Journal on Optimization*, 0(0). DOI: [10.1287/ijoo.2023.0029](#).
3. Swamy, R., D.M. King, **I.G. Ludden**, K.W. Dobbs, and S.H. Jacobson (2024). "A practical optimization framework for political redistricting: A case study in Arizona." *Socio-Economic Planning Sciences*, 92. DOI: [10.1016/j.seps.2024.101836](#).
4. Dobbs, K.W., R. Swamy, D.M. King, **I.G. Ludden**, and S.H. Jacobson (2024). "An Optimization Case Study in Analyzing Missouri Redistricting." *INFORMS Journal on Applied Analytics*, 54(2):162-187. DOI: [10.1287/inte.2022.0037](#).

5. **Ludden, I.G.**, D.M. King, and S.H. Jacobson (2023). "3D geo-graphs: Efficient flip verification for the spherical zoning problem." *Discrete Applied Mathematics*, 338:329-346. DOI: [10.1016/j.dam.2023.07.004](https://doi.org/10.1016/j.dam.2023.07.004).
6. **Ludden, I.G.**, R. Swamy, D.M. King, and S.H. Jacobson (2023). "A Bisection Protocol for Political Redistricting." *INFORMS Journal on Optimization*, 5(3):233-255. DOI: [10.1287/ijoo.2022.0084](https://doi.org/10.1287/ijoo.2022.0084).
7. **Ludden, I.G.**, S.H. Jacobson, and J.A. Jokela (2022). "Excess Deaths by Sex and Age Group in the First Two Years of the COVID-19 Pandemic in the United States." *Health Care Management Science*. DOI: [10.1007/s10729-022-09606-3](https://doi.org/10.1007/s10729-022-09606-3).
8. Pavlik, J.A., **I.G. Ludden**, and S.H. Jacobson (2021). "SARS-CoV-2 aerosol risk models for the Airplane Seating Assignment Problem." *J Air Trans Mgmt*, 99. DOI: [10.1016/j.jairtraman.2021.102175](https://doi.org/10.1016/j.jairtraman.2021.102175).
9. Pavlik, J.A., **I.G. Ludden**, S.H. Jacobson, and E.C. Sewell (2021). "Airplane Seating Assignment Problem." *Service Science*, 13(1):1-52. DOI: [10.1287/serv.2021.0269](https://doi.org/10.1287/serv.2021.0269).
10. **Ludden, I.G.**, A. Khatibi, D.M. King, and S.H. Jacobson (2020). "Models for Generating NCAA Men's Basketball Tournament Bracket Pools." *JQAS*, 16(1):1-15. DOI: [10.1515/jqas-2019-0022](https://doi.org/10.1515/jqas-2019-0022).

Peer-reviewed
Conf. Papers

11. Deshpande, S.P., **I.G. Ludden**, and S.H. Jacobson (2023). "Votemandering: Strategies and Fairness in Political Redistricting." *Third ACM conference on Equity and Access in Algorithms, Mechanisms, and Optimization (EAAMO'23)*.

Presentations

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|---|----------|
| INFORMS Optimization Society Conference (IOS) | Mar 2026 |
| – Session FA08 – Applications of Mixed-Integer Linear Programming | |
| – "Evaluating Redistricting Reform Proposals with MIP-based Heuristics" | |
| Franciscan University of Steubenville STEM Seminar (invited talk) | Mar 2025 |
| – "Algorithms, Incentives, and Autonomy: Improving Fairness while Respecting Free Will" | |
| INFORMS Optimization Society Conference (IOS) | Mar 2024 |
| – Session SunB3 – Topics in mixed integer programming | |
| – "A Bilevel Mixed-Integer Program for the Define-Combine Redistricting Procedure" | |
| Wabash College Math/CS Colloquium | Oct 2023 |
| – "Connected Recursive Bisection and Perfect Hierarchical Matchings" | |
| INFORMS Annual Meeting | Oct 2022 |
| – Session TD09 – PSOR Flash Session | |
| – "A Bilevel Define-Combine Formulation with Applications to Political Redistricting" | |
| INFORMS Computing Society Conference (ICS) | Jan 2022 |
| – Session MB4 – Network Applications | |
| – "Analyzing and Modeling the Define-Combine Procedure for Political Redistricting" | |
| INFORMS Annual Meeting | Oct 2021 |
| – Chair: Session WE25 – Combinatorial Optimization | |
| – "3-D Geo-graphs: Efficient Flip Verification for 3-D Graph Partitioning" | |
| INFORMS Annual Meeting | Oct 2019 |
| – Session WC43 – Political Redistricting | |
| – "A Bisection Protocol for Political Redistricting" | |

Mentoring	Daniel Leverett (RHIT) – Math capstone project: “Bayesian Inference of Player Roles in a Social Deduction Game” – Rose Research Fellow project on social deduction games	Dec 2023 – Present
	Mason Wiedow (RHIT) – Research assistant for computational redistricting project – Poster presentation at Spring 2026 Indiana Section MAA Meeting	Sep 2025 – Present
	Akaash Kolachina and Kylie Zhang (UIUC) – PURE Program project on redistricting visualization	Fall 2020
Funding	KEEN Seed Grant as Subaward from UIUC (\$5,000) – “Enhancing Learning of Object-Oriented Design Principles with GenAI-assisted Practice”	Oct 2025 – Present
	CSSE Summer Innovation Grant (\$5,000) – “Integration of GenAI in CSSE 220”	Jun-Aug 2025
	KEEN Summer Course Development Grant (\$7,000) – “CSSE 490: Algorithmic Game Theory”	Jun-Aug 2024
Outreach	Volunteer Judge for Congressional App Challenge , IN08	Nov 2025
	Volunteer Judge for St. Patrick School Science Fair	Jan 2025, Feb 2026
	Volunteer, Ryves Youth Center	Apr 2026 – Present
	Python Instructor for Connecting With Code 2024	Summer 2024
	Community Computer Lab Volunteer, Salt & Light – Supervise public computer lab of not-for-profit grocery and thrift store – Develop and deliver training program for REcompute refurbished laptop recipients	July 2021 – Mar 2023
Consulting	Project PRE.CISE – Organize eight-week program (workshops, panels) for NSF REU Supplement students – Objectives: build community; inform students of grad school and research careers – Collaboration between CRA-E , CERP , and NSF CISE directorate	Summer 2021
Industry Experience	Software Engineer, PilotFish Technology (Tampa, FL) – Integration platform development (Java, XML/XPath)	Jan–Aug 2017
	Computer Science Intern, LGS Innovations (Tampa, FL) – Communication system modeling (MATLAB)	Summer 2016
	Software Engineer Intern, Garmin Intl. (Olathe, KS) – Embedded development for GPS fitness watches (C)	Summer 2015
	Programming Intern, FitzMark, Inc. (Indianapolis, IN) – Dispatch application development	Summer 2014